First principles evidence of stiffening during spinodal decomposition in FeCr

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Abstract

We explain stiffening during spinodal decomposition in bcc δ-ferrite previously observed in cast austenitic stainless steels by combining experimental measurements with first-principles simulations of bcc $Fe_{1-x}Cr_x$. Quantum elastodynamic calculations predict surface acoustic wave (SAW) peak splitting arising from elastic stiffening of δ-ferrite during spinodal decomposition. Classical first-principles elastodynamics also quantitatively calculate Rayleigh SAW peak splitting arising from spinodal stiffening as a non-destructive indicator of embrittlement, in agreement with experiment. More broadly, similar SAW-based stiffening signatures may prove useful for detecting other embrittlement mechanisms in structural alloys.

Keywords: Stainless steel, binary alloy, iron-chrome, spinodal decomposition, surface acoustic wave, Ab initio calculation, stiffening, embrittlement, thermal diffusivity, δ–ferrite, quantum elastodynamics (QED)

Highlights

- Ab initio simulations predict stiffening during spinodal decomposition in δ–ferrite
- Composition-dependent elasticity maps to surface acoustic wave frequency shifts
- Theory and experiment agree: stiffening non-destructively indicates embrittlement

1. Introduction and Background

Recent structure–property studies have linked spinodal decomposition to stiffening of bcc δ–ferrite in cast austenitic stainless steels (CASS), and have shown that this stiffening can be detected through picosecond ultrasonic surface acoustic wave (SAW) peak splitting [1–3]. This paper establishes a quantitative theoretical foundation for predicting that SAW signature from first principles by combining elastodynamics with density functional theory (DFT) to connect composition, elastic response, and Rayleigh-wave observables.

For high–melting-point metals (including bcc δ–ferrite), elasticity is often closely tied to plastic response: dislocation energetics are largely stored in the Hookean lattice outside the dislocation core, and thermally activated processes during deformation can modify elastic properties [4]. More broadly, cohesive properties quantified by lattice elasticity correlate with melting point through their shared dependence on interatomic bonding and electronic structure [5–9]. In ferromagnetic Fe–Cr alloys, these quantities vary strongly with concentration [7], and the experimentally observed SAW peak splitting upon spinodal decomposition is consistent with an anomalous concentration dependence of elastic constants associated with electronic-structure effects (including Fermi-surface topology changes [8]).

During spinodal decomposition of bcc δ–ferrite, chromium partitioning produces Fe-rich α and Cr-rich α' regions whose differing compositions imply differing stiffness. As a result, elastic stiffening can serve as a

mechanistically grounded proxy for embrittlement in stainless steels undergoing thermal and/or irradiation-assisted aging [2,3,10–14]. Because the Rayleigh SAW frequency depends directly on elastic moduli, picosecond ultrasonic pump–probe spectroscopy provides a direct, non-destructive route to infer stiffness changes and therefore track the progression toward embrittlement [12].

Motivated by these links, we develop a thermodynamics-to-observable workflow for bcc $Fe_{1-x}Cr_x$ as a model for δ–ferrite: (i) we identify spinodally unstable composition windows using the curvature of the demixing free energy, (ii) we quantify the composition dependence of cubic elastic constants from DFT, and (iii) we map these elastic inputs into Rayleigh SAW frequency ranges using elastodynamic constraints relevant to picosecond ultrasonics. Finally, we separate a solid-solution trend $f_{\mathrm{SAW}}(x_{\mathrm{Cr}})$ from an evolution trend $f_{\mathrm{SAW}}(\phi)$ parameterized by the spinodal conversion fraction ϕ from our earlier atomistic simulations of decomposed Fe–Cr [1], enabling direct interpretation of SAW peak shifts and splitting in terms of composition and decomposition state.

2. Methods

2.1. Elastodynamic Methods

Elastic waves in each homogeneous alloy obey the elastodynamic wave equation [10]:

$$\rho \frac{\partial^2 u_i}{\partial t^2} = C_{ijkl} \frac{\partial^2 u_k}{\partial x_j \partial x_l} \tag{1}$$

where C_{ijkl} are fourth-order stiffness tensor entries indexed by $i, j, k, l = 1, 2, 3$, ρ is density, and u is the displacement field. On its own, Eq. 1 describes bulk waves in an unbounded medium. The SAW measured by transient grating spectroscopy (TGS), however, is a Rayleigh wave guided by the free surface. It is obtained from Eq. 1 for a half-space $x_3 \geq 0$ by requiring a traction-free surface, $\sigma_{i3} = C_{i3kl}\,\partial u_k/\partial x_l = 0$ at $x_3 = 0$, and displacements that decay with depth. Because crystal orientation is not resolved in the present calculations, we solve this surface problem for the isotropic Voigt–Reuss–Hill (VRH) average of each elastic tensor (Sec. 2.4). The Rayleigh velocity is then $v_R = v_s\sqrt{q}$, where q is the root in $0 < q < 1$ of

$$q^3 - 8q^2 + (24 - 16\beta)q - 16(1-\beta) = 0, \qquad \beta = v_s^2/v_p^2, \tag{2}$$

with $v_s = \sqrt{G/\rho} = \sqrt{E/[2(1+\nu)\rho]}$ and $v_p = \sqrt{(B + 4G/3)/\rho}$. In practice we evaluate $v_R = (0.874 + 0.196\nu - 0.042\nu^2 - 0.055\nu^3)\,v_s$, which reproduces the root of Eq. 2 to within 0.02% for $0.20 \leq \nu \leq 0.35$. The SAW frequency is $f_{\mathrm{SAW}} = v_R/\lambda$ with $\lambda = 6.4\,\mu$m, the grating period of our TGS measurements [1]. Density is interpolated linearly between bcc Fe (7870 kg m^{-3}) and Cr (7200 kg m^{-3}).

The elastic tensor is calculated using a variable cell relaxation algorithm that consists of a loop over iterative self-consistent field (SCF) calculations while adjusting atomic positions and lattice constants. Elasticity theory relates the strain applied to the unit cell to the resulting calculated stress as an inverse problem, which could be solved using a non-linear solver with prior data as an initial guess vector. Deterministic thermoelastic relationships are also used to calculate elastic tensors from changes in total energy and volume using ab initio stress calculations. Both methods are used in a self-consistent fashion to converge to a solution that solves the elasticity inverse problem, $\underline{C}\vec{\epsilon} = \vec{\sigma}$, as follows [15,16]:

$$\begin{bmatrix} C_{1111} & C_{1122} & C_{1133} & C_{1123} & C_{1113} & C_{1112} \\ & C_{2222} & C_{2233} & C_{2223} & C_{2213} & C_{2212} \\ & & C_{3333} & C_{3323} & C_{3313} & C_{3312} \\ & & & C_{2323} & C_{2313} & C_{2312} \\ & \mathit{symm} & & & C_{1313} & C_{1312} \\ & & & & & C_{1212} \end{bmatrix} \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ 2\epsilon_{23} \\ 2\epsilon_{13} \\ 2\epsilon_{12} \end{bmatrix} = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix} \tag{3}$$

or in Voigt notation:

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ & & C_{33} & C_{34} & C_{35} & C_{36} \\ & & & C_{44} & C_{45} & C_{46} \\ & \textit{symm} & & & C_{55} & C_{56} \\ & & & & & C_{66} \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \\ \epsilon_6 \end{bmatrix} = \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix} \tag{4}$$

2.2. DFT Methods

Material properties are calculated from a self-consistent, non-linear field calculation (Fig. S10), a variational method that optimizes the charge density and adjusts atomic positions until a threshold in energy minimization is achieved in a simplified ion-core model of atoms and electrons [17].

Self-consistent, non-linear field calculation algorithms calculate external potentials with a Poisson equation solver, (Fig. S10) which depends on simulation geometry, atomic positions of ions, pseudopotentials used for the atoms involved, and other inputs. This external potential is then used as a guess for the electron density with a Schrödinger equation solver [18], which is then used as inputs to a self-consistent field loop that calculates the charge density iteratively to minimize the total energy. The accuracy of this algorithm depends on the cut-off applied to the energy minimization, the charge density, the size of the momentum grid, and other convergence thresholds on the outputs [17–19].

In order to model the BCC Fe–Cr δ–ferrite solid solution, the special quasirandom structure (SQS) methodology was employed [20]. The mcsqs software package within the Alloy Theoretic Automated Toolkit (ATAT) was used in order to generate SQS supercells. The Vienna ab initio Simulation Package (VASP) [21] was used to perform the density functional theory (DFT) total energy calculations, which were used to derive the elastic properties as well as the enthalpy of mixing. Spin–polarized DFT calculations were performed on 40–atom SQS supercells employing Γ–point sampling of reciprocal space along with a 500 eV planewave cutoff energy. The Projector–Augmented Wave (PAW) method [22] was employed for the ionic pseudopotential, and the Perdew–Burke–Ernzerhof parameterization [23] of the Generalized Gradient Approximation (GGA) of the exchange–correlation functional. Calculations were performed at $x_{\mathrm{Cr}} = 0$, 0.20, 0.50, 0.95, and 1.00.

2.3. Atomistic simulations of decomposed Fe–Cr

The spinodal conversion fraction ϕ and the corresponding elastic moduli are taken from our earlier atomistic study [1] (Sec. 3.7 and Fig. 8 therein). These simulations were performed with LAMMPS [24], not VASP, and include no electronic structure. A 50×50×50 bcc cell (lattice parameter 2.866 Å) of 250,000 atoms with equiatomic composition was populated randomly and phase-separated by Monte Carlo atom swaps at 10 K (3000 attempts every 50 steps for 40,000 steps). The swaps were driven by a structure-generation pair interaction only: zero for like pairs and a strongly repulsive nearest-neighbour Fe–Cr Lennard-Jones term ($\epsilon = 10$ eV). Intermediate states were produced by converting a fraction (10–50%) of the Fe atoms to Cr and the same fraction of Cr atoms to Fe. After box and ionic relaxation, the elastic constants of each state were calculated by finite deformation (strain 10^{-6}) with the Fe–Cr two-band embedded-atom-method (EAM) potential of Bonny et al. [25], for ten alloys generated with different random seeds. On this scale, $\phi = 0.5$ is the random homogeneous alloy and $\phi = 1$ the fully separated structure. Relaxed cell volumes were not retained, so the density is held at the equiatomic value of 7535 kg m^{-3} when these moduli are mapped to f_{SAW} with Eq. 2. All LAMMPS inputs, random seeds, and potential files are provided in the accompanying repository. For direct comparison with the homogeneous-alloy DFT and literature data, we also computed random (homogeneous) bcc alloys at $x_{\mathrm{Cr}} = 0$, 0.20, 0.50, 0.95, and 1.00 with the same stiffness procedure, cell, and potential, using ten random seeds for each mixed composition (one for the pure elements), and at $x_{\mathrm{Cr}} = 0.10$, 0.30, 0.36, 0.40, 0.60, 0.70, and 0.80 with three seeds each. The random $Fe_{50}Cr_{50}$ alloy reproduces the published value ($C_{11} = 252.9 \pm 0.1$ versus 252.7 ± 0.1 GPa), and the seed-to-seed scatter in C_{11} is at most 0.25 GPa. These runs use the current NIST release of the Bonny et al. potential files, which are numerically identical to those used in Ref. [1] apart from two format corrections required by current LAMMPS versions: zero-filled pair tables in the s-band file and finite values at $r = 0$.

To extend the decomposition series beyond the equiatomic alloy, we repeated the full procedure at $x_{Cr} = 0.20$, 0.36, and 0.95 in a $25\times25\times25$ cell (31,250 atoms) with the same number of attempted swaps per atom. Away from 50/50, converting equal fractions of the Fe and Cr atoms changes the overall composition. We therefore converted a fraction c of the Cr atoms and a fraction cN_{Cr}/N_{Fe} of the Fe atoms, with $c = 2(1 - x_{Cr})(1 - \phi)$; this preserves the composition (to within 0.004 in all structures) and is identical to the original procedure at 50/50. One alloy was generated per composition. As a check, the same small-cell procedure at 50/50 reproduces C_{11} of the original series to within 0.8 GPa at every ϕ.

2.4. Analysis of DFT Results

With the elastic tensor obtained from quantum mechanical calculations, the isotropic VRH moduli used in Eq. 2 are calculated as follows; the same moduli enter the Pugh embrittlement criterion and the thermal estimates reported in the Supplementary Material.

$$\begin{aligned} C' &= \frac{C_{11} - C_{12}}{2}, & B &= \frac{C_{11} + 2C_{12}}{3}, \\ C_{11} &= \frac{3B + 4C'}{3}, & C_{12} &= C_{11} - 2C', \\ G_V &= \frac{C_{11} - C_{12} + 3C_{44}}{5}, & G_R &= \frac{5(C_{11} - C_{12})C_{44}}{4C_{44} + 3(C_{11} - C_{12})}, \\ G &= \frac{G_V + G_R}{2}, & E &= \frac{9BG}{3B + G}, & \nu &= \frac{3B - E}{6B}. \end{aligned} \tag{5}$$

Equation 5 gives the cubic forms used for the literature tensors. For the DFT points, B and G are VRH averages of the full calculated tensor, because the SQS cells are not exactly cubic. All figures are generated from tabulated inputs by scripts provided in the accompanying repository (see Data Availability).

3. Results

3.1. Spinodal stability from demixing free-energy curvature

The local stability of a binary solid solution is governed by the curvature of its free energy of mixing with respect to composition. Where $\partial^2\Delta G_{mix}/\partial x_{Cr}^2 < 0$, infinitesimal composition fluctuations lower the free energy, and demixing proceeds spontaneously without the need for nucleation barriers. Figure 1 is evaluated at 298.15 K from the concentration-dependent interaction models that Levesque et al. fitted to PWSCF–PAW and SIESTA–NC calculations [26], $\Omega(x) = (x - a)(bx^2 + cx + d)$, together with the ideal configurational entropy S_{mix}. Following the sign convention of that model, we plot the demixing free energy

$$\Delta G_{demix} = \Omega(x)\, x(1 - x) + TS_{mix} = -\Delta G_{mix}. \tag{6}$$

The spinodal criterion $\partial^2\Delta G_{mix}/\partial x_{Cr}^2 < 0$ therefore appears as positive curvature, $\partial^2\Delta G_{demix}/\partial x_{Cr}^2 > 0$, in Fig. 1b. Figure 1b compares three estimates of this curvature for the PWSCF–PAW model: the exact analytic second derivative, three-point numerical second differences of its tabulated values on the non-uniform composition grid, and a cruder two-point scheme together with a quadratic fit to its values. The three-point values fall on the exact curve. The two-point scheme distorts the magnitudes on the uneven grid, most strongly at $x_{Cr} = 0.70$, next to a duplicated composition in the tabulation, but agrees on where the instability begins.

Figure 1b shows that the homogeneous alloy is unstable for $0.21 \lesssim x_{Cr} \lesssim 0.75$ in the PWSCF–PAW model (with a second, narrow window at 0.87–0.98); the SIESTA–NC model gives $0.25 \lesssim x_{Cr} \lesssim 0.99$. Both windows contain the Fe–36Cr alloy in which spinodal decomposition was recently detected through TGS stiffening [30]. Because the entropic term lowers the curvature in proportion to T, these windows narrow at the higher temperatures used for aging. Within them, elastic and acoustic properties cannot be interpreted assuming a homogeneous solid solution. Instead, any observable elastic response must be understood as emerging from a system that is intrinsically driven toward compositional modulation, providing the foundation for linking free-energy curvature to stiffness evolution and ultimately to ultrasonic observables.

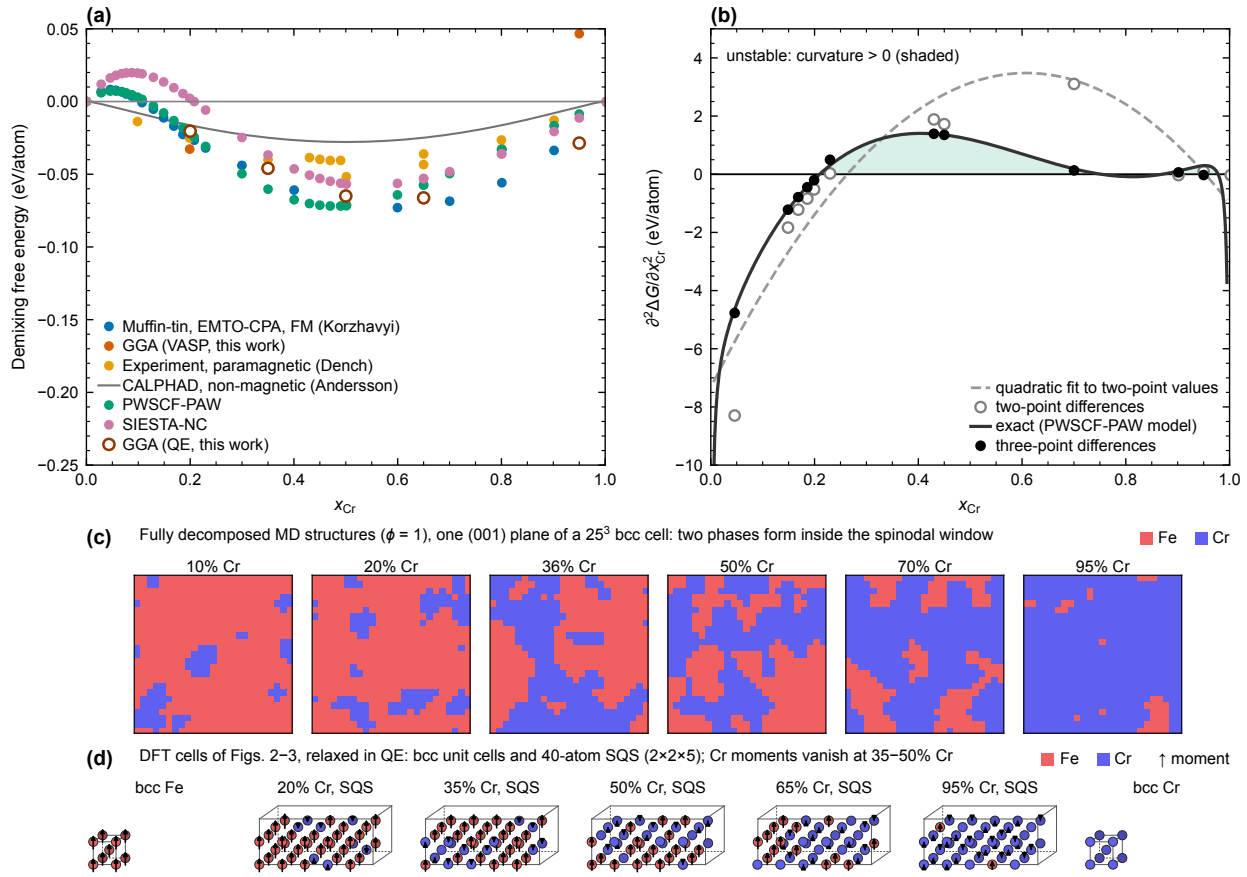


Figure 1: (a) Demixing free energy $\Delta G_{\mathrm{demix}} = -\Delta G_{\mathrm{mix}}$ (Eq. 6) versus chromium composition x_{Cr} at 298.15 K: muffin-tin (EMTO-CPA, ferromagnetic) enthalpies of Korzhavyi et al. [7], present GGA calculations with VASP (filled) and Quantum ESPRESSO (open circles; 40-atom SQS, references FM Fe and AFM Cr), calorimetric enthalpies of Dench [27] (measured near 1600 K relative to paramagnetic Fe and Cr), the CALPHAD bcc description of Andersson and Sundman [28] (non-magnetic part, line), and the PWSCF–PAW and SIESTA–NC interaction models of Levesque et al. [26]. All series include the ideal configurational entropy as in Eq. 6; the measured vibrational entropy of mixing, about $0.2\,k_{\mathrm{B}}$/atom [29], would change $\Delta G_{\mathrm{demix}}$ by at most 5 meV/atom at this temperature. (b) Curvature of the PWSCF–PAW demixing free energy. Solid line: exact analytic second derivative of the model. Filled circles: three-point second differences of its tabulated values on the non-uniform composition grid. Open circles: two-point second differences of the same values, with a quadratic least-squares fit (dashed). Shading marks positive demixing curvature, i.e., negative mixing curvature, which identifies spinodally unstable compositions. (c) One (001) atomic plane of fully decomposed ($\phi = 1$) MD structures (25^3 bcc cells; Sec. 2.3) at the indicated Cr contents; red, Fe; blue, Cr. Inside the spinodal window the alloy separates into interpenetrating Fe-rich α and Cr-rich α' domains, whereas at the dilute ends the minority element forms only isolated clusters. (d) The DFT cells behind the first-principles tensors of Figs. 2 and 3: bcc unit cells of Fe and Cr and 40-atom SQS ($2 \times 2 \times 5$ conventional cells), relaxed in Quantum ESPRESSO, with their local magnetic moments (arrows; red, Fe; blue, Cr). The Cr moments, antiparallel to Fe at 20% Cr, vanish between 35 and 50% Cr.

3.2. Elastic stiffening and elastodynamic mapping to Rayleigh SAW observables

To connect thermodynamic instability to experimentally measurable quantities, we compute the cubic elastic tensor across x_{Cr} and propagate these elastic inputs through the surface boundary-value problem of Sec. 2.1. Although f_{SAW} depends on the full tensor through B and G, we use the longitudinal stiffness C_{11} as a compact index of composition-dependent stiffening (Fig. 2a). The black circles are literature calculations by Razumovskiy et al. [8], the red diamonds are ours, and the green squares are MD (EAM) random alloys computed here with the procedure of our earlier study [1]; the open squares are the corresponding fully decomposed structures, which are stiffer than the random alloy at every composition, most strongly at intermediate compositions (e.g., 257.4 versus 247.3 GPa at $x_{\mathrm{Cr}} = 0.36$). The present DFT tensors are

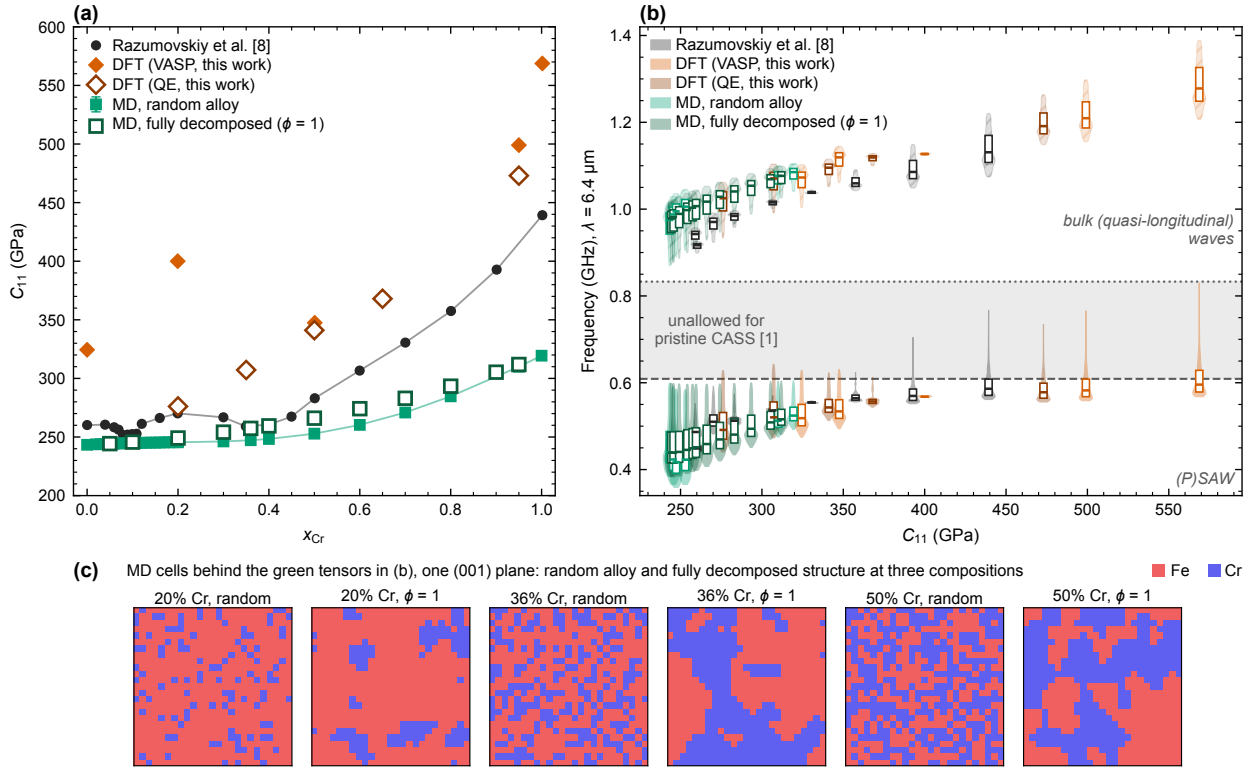


Figure 2: (a) C_{11} of bcc $Fe_{1-x}Cr_x$: literature calculations of Razumovskiy et al. [8] (black circles; reconstructed from digitized C', C_{44}, and B), present DFT calculations on 40-atom SQS with VASP (filled red diamonds) and Quantum ESPRESSO (open diamonds; SM S2), MD (EAM) random alloys (green squares; mean ± standard deviation of ten alloys at the DFT compositions and three elsewhere; Sec. 2.3), and fully decomposed MD structures ($\phi = 1$, open squares; 25^3 cells). (b) Orientation-resolved acoustic frequencies at $\lambda = 6.4\,\mu$m, plotted at the C_{11} of each tensor in (a). For each tensor, 10,000 random grain orientations were computed with the surface Green's-function method of Du and Zhao [10], as in Ref. [1]. Lower violins: (pseudo-)SAW frequencies (pseudo-SAW retained when its intensity is at least 2×10^{-3} of the SAW). Upper violins: quasi-longitudinal bulk waves along the same propagation directions. Boxes mark the interquartile range and median. The shaded band is the elastodynamically unallowed range of pristine 304-type CASS [1], from its maximum (pseudo-)SAW frequency (0.609 GHz, dashed) to its bulk-wave frequency (0.833 GHz, dotted). (c) One (001) atomic plane of the MD cells behind the green tensors: random alloy and fully decomposed structure at $x_{Cr} = 0.20$, 0.36, and 0.50 (red, Fe; blue, Cr).

systematically stiffer than the reference data (e.g., $C_{11} = 324$ versus 260 GPa for bcc Fe) and correspondingly predict higher frequencies. Possible contributors include the Γ-point-only k-point sampling, the finite SQS size, and the magnetic configuration. We therefore interpret relative trends rather than absolute frequencies. Independent Quantum ESPRESSO calculations on 40-atom SQS with converged Brillouin-zone sampling (open diamonds; SM S2) agree with the literature at $x_{Cr} = 0.20$ and 0.95 ($f_{SAW} = 0.509$ and 0.591 GHz versus 0.512 and 0.590 GHz) but rise monotonically in between, without the minimum near 36% Cr.

As shown in Fig. 2, C_{11} depends systematically on chromium concentration (non-monotonically in the literature and DFT data), producing corresponding shifts in the Rayleigh SAW frequency. For a homogeneous solid solution, each composition corresponds to a single tensor, and hence to a single isotropic f_{SAW} but a distribution of (P)SAW frequencies over grain orientations. In our earlier study, elastodynamic calculations for pristine CASS over randomly sampled grain orientations defined an "unallowed" frequency range, above the highest (pseudo-)SAW frequency and below the bulk-wave frequency, in which no surface mode of the pristine material exists; (P)SAW peaks observed there after aging were attributed to stiffened δ-ferrite [1]. Figure 2b repeats that calculation for every Fe–Cr tensor (10,000 grain orientations each, $\lambda = 6.4\,\mu$m) [10]. The bulk waves of all tensors lie above the range (≥ 0.87 GHz). In the (P)SAW branch, no literature tensor up to $x_{Cr} = 0.7$ and none of the MD tensors reaches it. Full decomposition raises the highest MD

(P)SAW frequency, most at intermediate compositions (from 585 to 596 MHz at $x_{Cr} = 0.36$), but not into the range. Only the stiffest tensors enter it: Cr-rich literature alloys (9% and 29% of grains above 0.609 GHz at $x_{Cr} = 0.9$ and 1.0) and the present DFT tensors (30% and 46% at $x_{Cr} = 0.95$ and 1.0; their systematic overstiffening also places 6–8% of grains of Fe and $Fe_{50}Cr_{50}$ in the range). Frequencies in this range therefore indicate stiffness beyond that of a homogeneous solid solution of δ-ferrite or Fe–36Cr composition, consistent with the stiffer Cr-rich α' regions formed by spinodal decomposition (Fig. 3a). Within the spinodally unstable composition window, elastic stiffening therefore translates into measurable acoustic signatures, enabling stiffness changes to be inferred non-destructively.

3.3. Separating composition trends from spinodal conversion trends

While elastic constants vary with chromium concentration even in a homogeneous solid solution, spinodal decomposition introduces an additional, time-dependent microstructural evolution. To disentangle these effects, it is necessary to distinguish changes driven purely by composition from those arising from the progression of phase separation. We therefore report two complementary relationships that isolate these contributions within a unified elastodynamic framework.

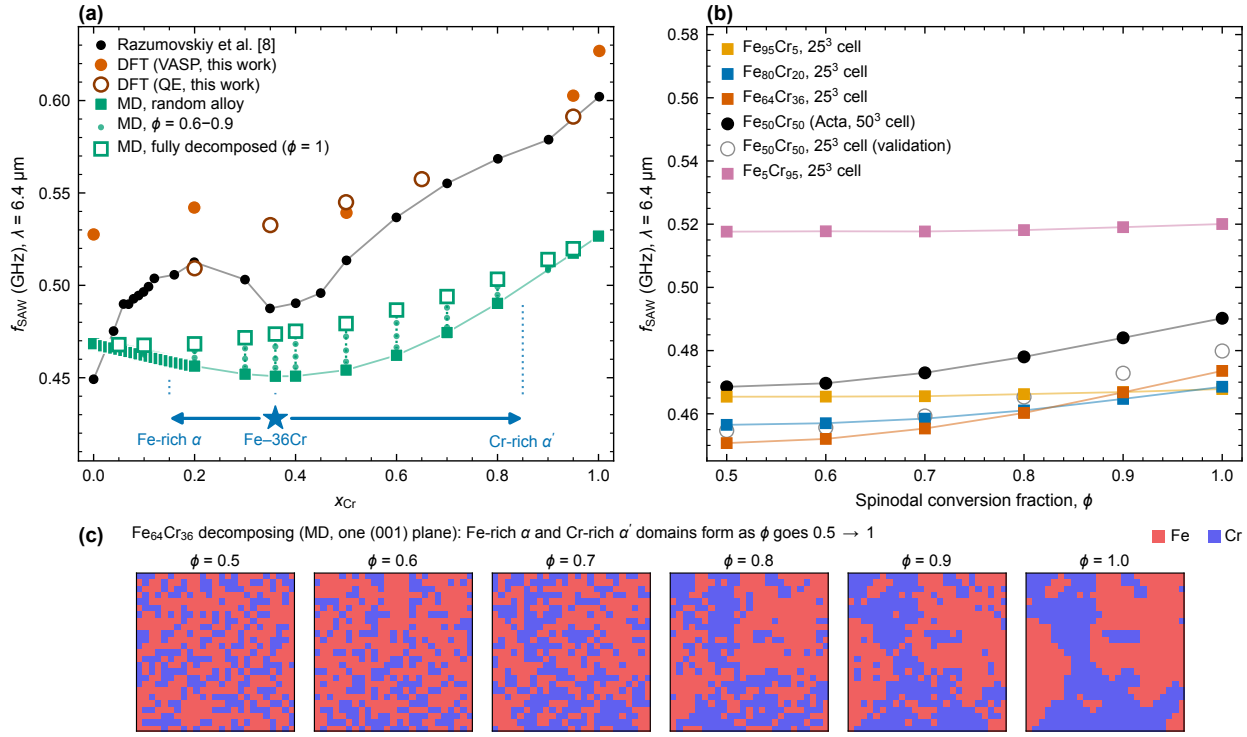


Figure 3: (a) Rayleigh SAW frequency f_{SAW} versus chromium composition x_{Cr} ($\lambda = 6.4\,\mu$m) for the literature (black), present DFT (VASP, filled red; Quantum ESPRESSO, open), and MD random-alloy (green) data of Fig. 2. At each MD composition, small dots ($\phi = 0.6$–0.9) and an open square ($\phi = 1$) show the frequency of the decomposing structure, each series anchored to the random-alloy value at that composition, so the ladder height is the stiffening on decomposition (as in b). Below the data, the star marks Fe–36Cr [30] and the arrows indicate schematically the partitioning toward Fe-rich α and Cr-rich α' during spinodal decomposition; dotted lines rise from each composition toward the data. Arrow endpoints are illustrative, not measured phase compositions. (b) f_{SAW} versus spinodal conversion fraction ϕ from the MD (EAM) decomposition simulations (Sec. 2.3): equiatomic $Fe_{50}Cr_{50}$ from Ref. [1] (black, 50^3 cell, fixed density 7535 kg m^{-3}), and $Fe_{95}Cr_5$, $Fe_{80}Cr_{20}$, $Fe_{64}Cr_{36}$, and Fe_5Cr_{95} from the present 25^3-cell calculations (squares); open circles repeat $Fe_{50}Cr_{50}$ in the 25^3 cell as a check. $\phi = 0.5$ is the random alloy and $\phi = 1$ the fully separated structure. (c) One (001) atomic plane of the 25^3 MD cell of $Fe_{64}Cr_{36}$ from $\phi = 0.5$ to 1.0 (red, Fe; blue, Cr): Fe-rich α and Cr-rich α' domains form and coarsen as ϕ increases; these are the structures whose frequencies are plotted in (b).

Figure 3a shows how chromium partitioning produces stiffening. The Fe–36Cr composition lies near a local minimum of the homogeneous $f_{SAW}(x_{Cr})$ curve. Spinodal decomposition moves material toward Fe-rich

α and Cr-rich α' compositions (arrows); in both directions the homogeneous-alloy frequency increases. The DFT set, with only five compositions, is consistent with this non-monotonic dependence but cannot resolve the minimum. The MD random alloys, an independent classical model, also show a frequency minimum, at $x_{Cr} \approx 0.36$–0.40, i.e. at the Fe–36Cr composition, even though their C_{11} increases monotonically (Fig. 2a): the minimum comes from the shear moduli, which control the Rayleigh velocity (C_{44} decreases from 116 GPa for Fe to 106 GPa at $x_{Cr} = 0.5$). C_{11} is therefore a convenient index of stiffening, but the SAW response is governed by shear stiffness. Figure 3a describes the properties of the constituents; the effective response of the two-phase microstructure additionally depends on phase fractions and on the domain size relative to the SAW wavelength.

Figure 3b isolates the effect of decomposition at fixed overall composition ($Fe_{50}Cr_{50}$). In the MD (EAM) simulations, f_{SAW} increases monotonically from 0.469 GHz for the random alloy ($\phi = 0.5$) to 0.490 GHz for the fully separated structure ($\phi = 1$), an increase of about 4.6% under the fixed-density assumption. The empirical-potential and first-principles results therefore agree in sign. Repeating the decomposition at twelve compositions from $x_{Cr} = 0.05$ to 0.95 (Sec. 2.3) gives frequency increases between the random and fully separated states that are nearly symmetric about the equiatomic composition: 0.5%, 1.1%, 2.6%, 4.4%, 5.1%, 5.4%, and 5.5% for $x_{Cr} = 0.05$, 0.10, 0.20, 0.30, 0.36, 0.40, and 0.50, and 5.3%, 4.1%, 2.7%, 1.1%, and 0.5% for $x_{Cr} = 0.60$, 0.70, 0.80, 0.90, and 0.95. Stiffening on decomposition is therefore largest within the spinodally unstable window of Fig. 1b, including the Fe–36Cr composition, and nearly vanishes for the dilute alloys at either end. The fully separated state is an idealized limit, so these values bound the stiffening from above. At 50/50, the small-cell series lies about 2% below the original series in absolute frequency because the original moduli used a different isotropic average; the relative changes agree to within one percentage point. Together, $f_{SAW}(x_{Cr})$ captures the baseline elastic response of a homogeneous alloy, and $f_{SAW}(\phi)$ reflects the additional stiffening associated with progressive spinodal conversion. This separation enables experimental SAW measurements to be interpreted in terms of both composition and decomposition state, providing a pathway to distinguish early-stage spinodal evolution from equilibrium compositional effects.

4. Discussion

Taken together, the results establish a closed thermodynamics-to-observable pathway for interpreting picosecond ultrasonic measurements in δ–ferrite–containing stainless steels. The free-energy curvature criterion identifies the composition window in which the homogeneous solid solution is intrinsically unstable, defining the regime where elastic and acoustic properties must reflect ongoing or incipient spinodal decomposition rather than equilibrium alloy behavior.

Within this thermodynamically unstable regime, first-principles elasticity reveals systematic stiffening trends that are propagated through elastodynamic constraints into Rayleigh SAW frequency shifts. The direct mapping between elastic constants and SAW observables provides a quantitative basis for interpreting peak shifts and splitting as signatures of stiffness evolution, rather than purely phenomenological indicators.

Crucially, separating the solid-solution trend $f_{SAW}(x_{Cr})$ from the spinodal conversion trend $f_{SAW}(\phi)$ resolves a key ambiguity in experimental interpretation. While composition alone can shift SAW frequencies, progressive spinodal decomposition introduces additional stiffening that cannot be captured by composition-dependent elasticity alone. The combined framework therefore enables SAW peak splitting to be interpreted as a mechanistically grounded indicator of microstructural evolution toward embrittlement, rather than as a purely compositional effect.

This separation clarifies why SAW peak splitting emerges as a robust, non-destructive diagnostic of spinodal decomposition in CASS and related alloys, and suggests a general strategy for extending ultrasonic embrittlement detection to other phase-separating structural materials.

5. Conclusions

We presented a compact thermodynamics-to-elastodynamics workflow for bcc Fe–Cr δ–ferrite that links demixing free-energy curvature, first-principles elastic constants, and Rayleigh SAW observables. Free-

energy curvature provides a direct stability criterion that identifies spinodally unstable composition windows. DFT-derived cubic elasticity quantifies composition-dependent stiffening trends and supplies inputs for elastodynamic prediction. Combining these elements yields experimentally actionable relationships for interpreting picosecond ultrasonics across both composition and decomposition-state variables.

Data Availability

The tabulated data, the atomistic and first-principles simulation inputs and outputs, and the scripts that generate all figures will be made publicly available at https://github.com/saleemaldajani/elastopya, with an archived DOI.

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Supplementary Material

S1. Thermal transport estimates

Thermal properties in Fig. S16 are estimated with the Debye model, and a lower limit for the thermal conductivity is obtained at high temperature [31,32]:

$$\kappa = \frac{1}{3} C_v v_m \Lambda \tag{7}$$

$$\kappa_{min} \rightarrow k_B v_m \Lambda \rightarrow k_B v_m \Lambda_{min} = A k_B \left(\frac{N_A m}{M}\right)^{\frac{2}{3}} \rho^{\frac{1}{6}} E^{\frac{1}{2}} \tag{8}$$

$$\Theta_D = \frac{2\hbar v_m}{a_0 k_B} \rightarrow B_D \frac{\hbar}{k_B} \left(\frac{N_A m}{M}\right)^{\frac{1}{3}} \frac{E^{\frac{1}{2}}}{\rho^{\frac{1}{6}}} \tag{9}$$

where κ is thermal conductivity, Θ_D is the Debye temperature, C_v is the heat capacity at constant volume, v_m is the average acoustic velocity, Λ is the phonon mean free path, m is the number of atoms, M is the mass, ρ is the density, and E is Young's modulus. The constants A and B_D arise from approximating v_m as proportional to $\sqrt{E/\rho}$; we use $A = 0.87$ following Clarke [31]. A temperature-dependent alternative is the Cahill–Watson–Pohl model, $\kappa_{min}(T) = (\frac{\pi}{6})^{\frac{1}{3}} k_B n^{\frac{2}{3}} \sum_i v_i (\frac{T}{\Theta_i})^2 \int_0^{\Theta_i/T} \frac{x^3 e^x}{(e^x-1)^2}\, dx$ with $\Theta_i = v_i(\hbar/k_B)(6\pi^2 n)^{\frac{1}{3}}$ [33].

S2. First-principles calculations with Quantum ESPRESSO

Methods.. An independent set of first-principles calculations was performed with Quantum ESPRESSO 7.5 [34,35] on the MIT Engaging cluster, using PBE [23] and PAW potentials from pslibrary (Fe 3s3p3d4s, Cr 3s3p3d4s), cutoffs of 60/480 Ry (wavefunctions/density), and Marzari–Vanderbilt smearing of 0.02 Ry. The alloys are 40-atom SQS ($2 \times 2 \times 5$ conventional bcc cells; pair correlations optimised over three neighbour shells) at $x_{\mathrm{Cr}} = 0.20$, 0.35, 0.50, 0.65 and 0.95, sampled with a $4 \times 4 \times 2$ k-mesh. Starting moments are ferromagnetic, and each cell is fully relaxed (box lengths and ions). Elastic constants follow from the stress response to strains $\varepsilon_{xx} = \pm 0.5\%$ and $\gamma_{yz} = \pm 0.5\%$ with ions relaxed at each strain (central differences). Pure Fe (FM) and Cr (NM and commensurate AFM) use 2-atom cells with 16^3 meshes. Convergence for bcc Fe: C_{ij} change by ≤ 2 GPa between 60 and 80 Ry and are converged by a 12^3 mesh. Electronic structure: atom- and spin-projected DOS from a non-self-consistent $6 \times 6 \times 3$ run with the optimised tetrahedron method; Fermi-surface sections on the $k_z = 0$ plane; band structures along Γ–H–N–Γ–P–H. Phonons: density-functional perturbation theory on a 6^3 q-grid, Fourier-interpolated along the same path.

To be added when the runs finish: unfolded Fermi surfaces and band structures of Fe-rich SQS ($x_{\mathrm{Cr}} = 0$, 0.05, 0.10); DOS at $x_{\mathrm{Cr}} = 0.20$ and 0.80; Fe–80Cr elastic constants; and the full set repeated at the room-temperature experimental lattice parameter, including new SQS at $x_{\mathrm{Cr}} = 0.30$, 0.40 and 0.45.

S3. Proposed mechanism figures

The two figures below collect the mechanism behind the anomalous elasticity and the stiffening on decomposition (Fig. SS8), and its consequence for embrittlement (Fig. SS9). They are placed here for now; we may move them to the main text.

S4. Additional figures

The following figures are retained from the original draft for completeness and traceability.

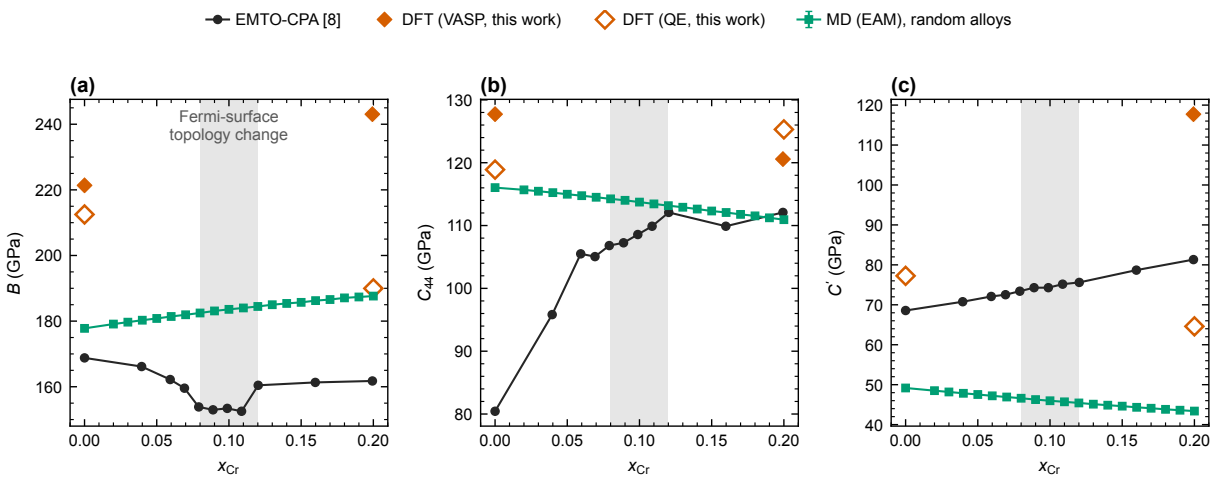


Figure S1: Fe-rich elastic constants ($x_{\mathrm{Cr}} \leq 0.2$): bulk modulus B, C_{44} and C' from EMTO-CPA [8], MD (EAM) random alloys on a 1% composition grid, and the present VASP and Quantum ESPRESSO calculations. The shaded band (0.08–0.12) marks the composition range of the Fermi-surface topology change reported in Ref. [8]; the classical potential has no electronic structure and does not reproduce it.

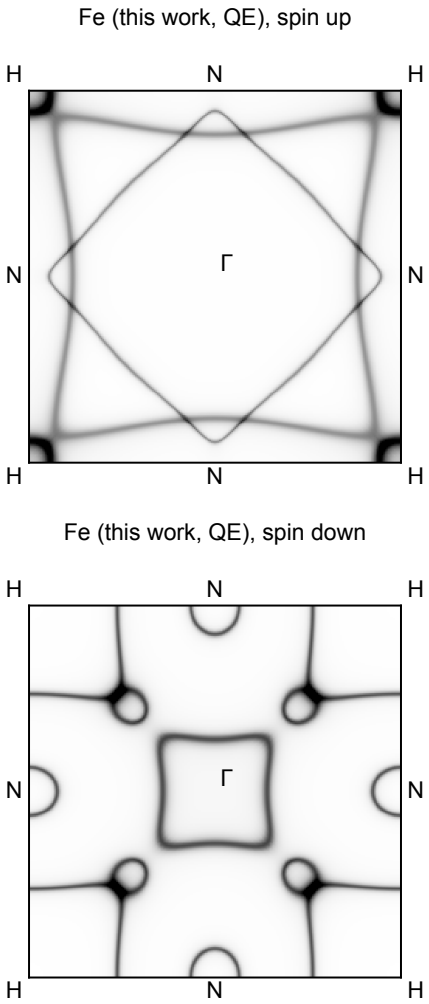


Figure S2: Fermi surface of ferromagnetic bcc Fe in the (001) plane through Γ (square with H at the corners), spin up and spin down, shown as a broadened spectral function in the layout of Korzhavyi et al. [7] (their Fig. 6).

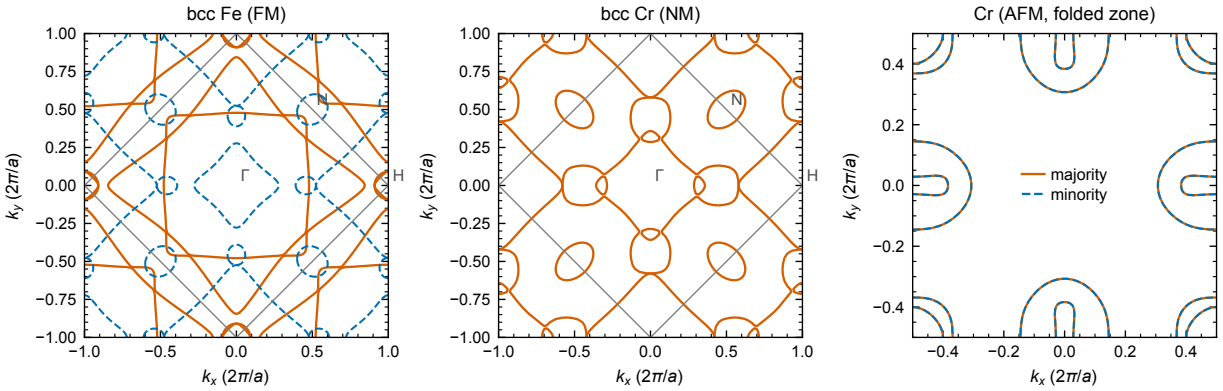


Figure S3: Fermi-surface sections ($k_z = 0$) of bcc Fe (FM; solid, majority; dashed, minority), bcc Cr (NM) and commensurate AFM Cr (folded simple-cubic zone; both spin channels coincide). The nested electron and hole sheets of Cr are responsible for its incommensurate spin-density wave.

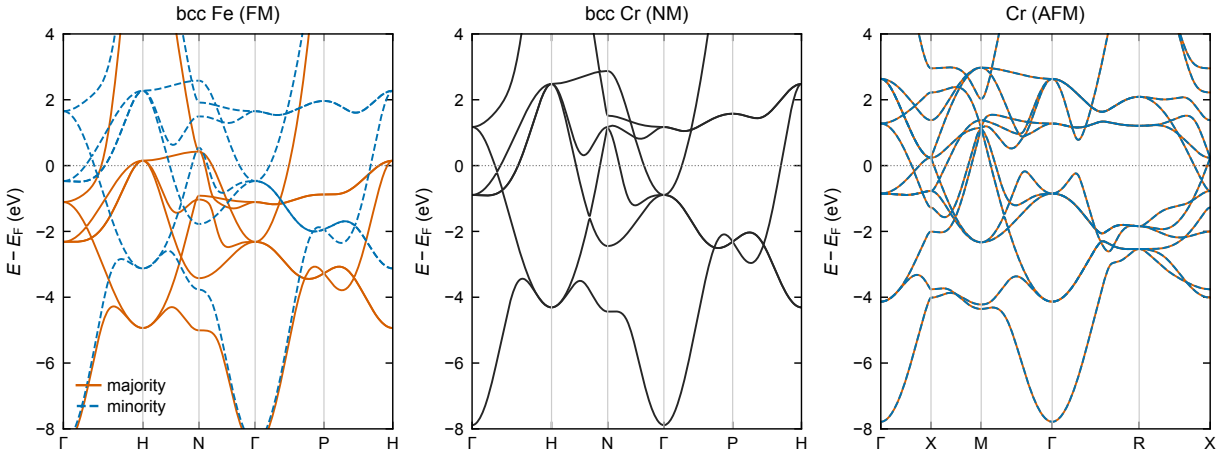


Figure S4: Electronic band structures of bcc Fe (FM), bcc Cr (NM) and AFM Cr (simple-cubic path), Quantum ESPRESSO.

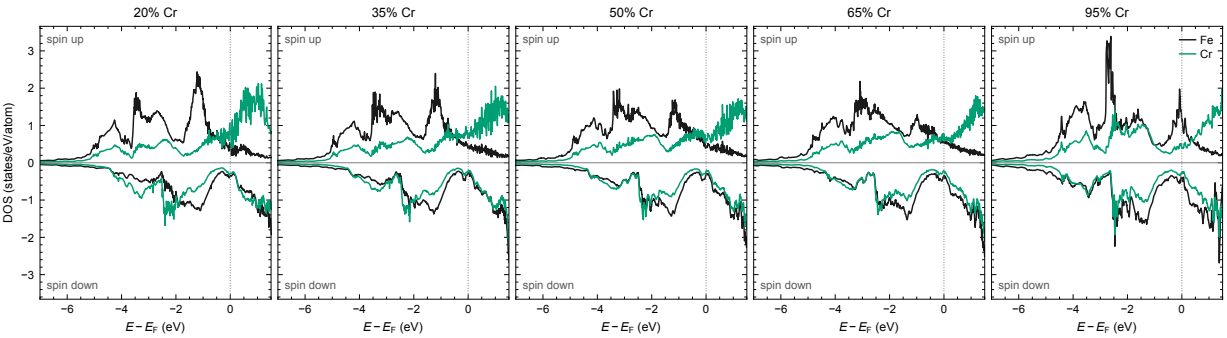


Figure S5: Fe- and Cr-projected densities of states per atom of the relaxed 40-atom SQS (spin up above, spin down below the axis), in the layout of Korzhavyi et al. [7] (their Fig. 4).

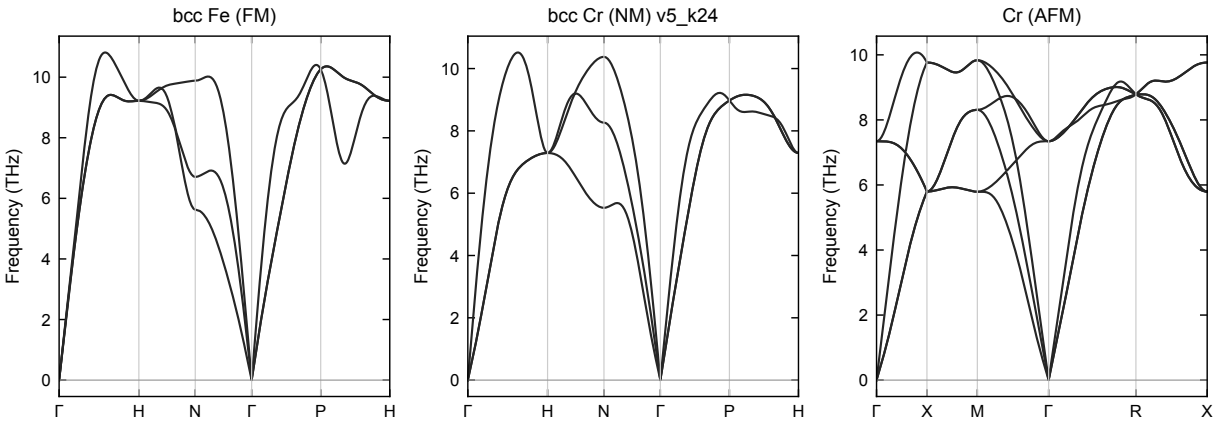


Figure S6: Phonon dispersions (DFPT) of bcc Fe (FM, $k = 16^3$), bcc Cr (NM, $k = 24^3$) and AFM Cr ($k = 12^3$, simple-cubic path). All frequencies are real.

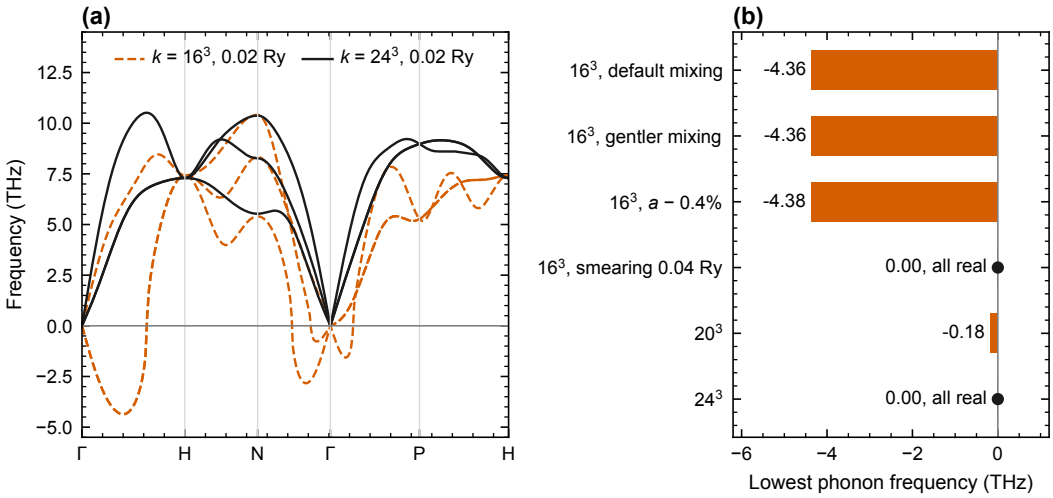


Figure S7: Brillouin-zone sampling of the NM Cr phonons. (a) Dispersion at an electronic k-mesh of 16^3 (imaginary branches plotted below zero) and 24^3 (all real). (b) Lowest frequency along the path for each setting: SCF mixing and a 0.4% change of the lattice parameter have no effect, whereas a denser mesh or wider smearing removes the imaginary modes. They are an artefact of under-sampling the nested Fermi surface of Cr, not a dynamical instability.

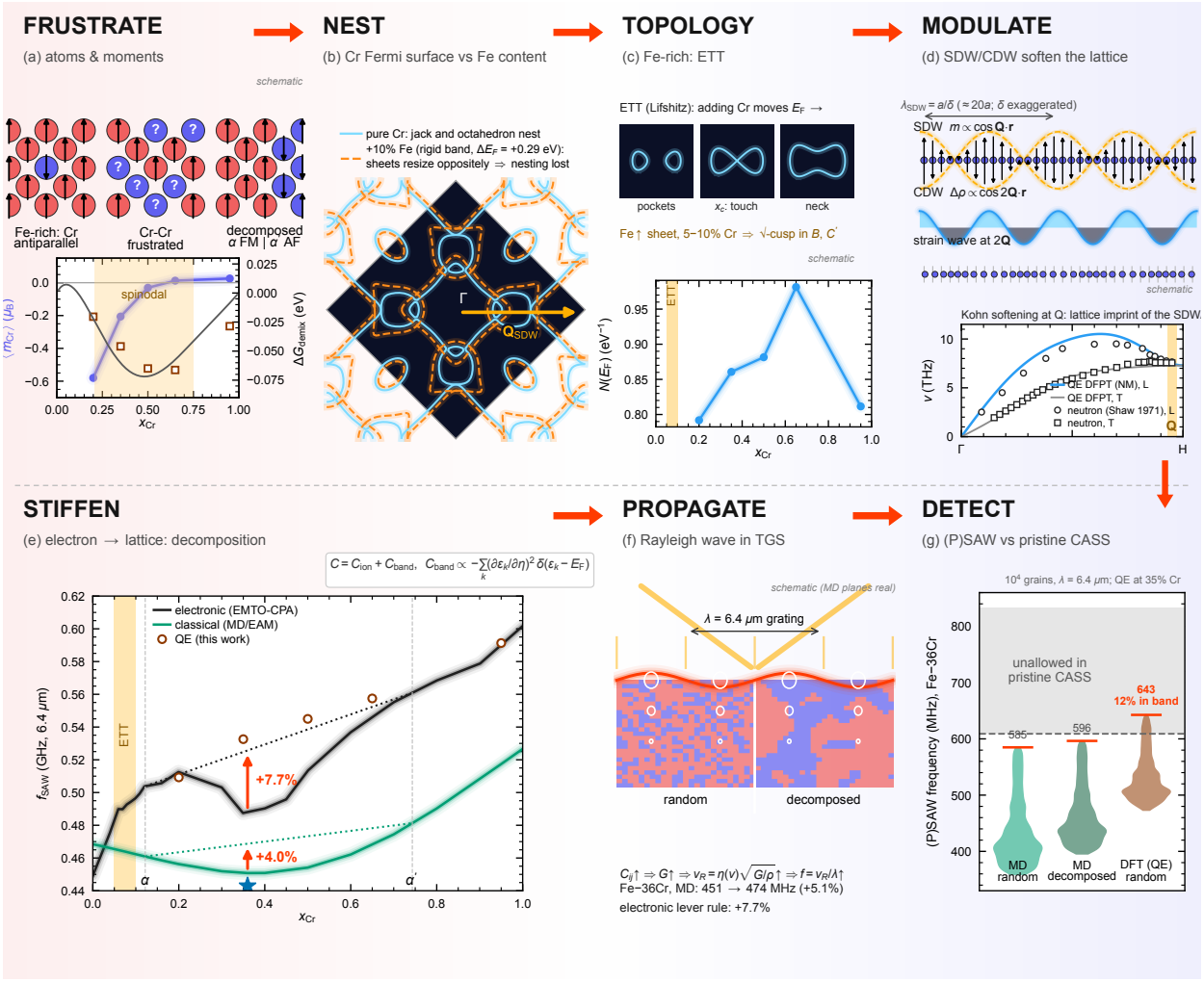


Figure S8: Mechanism of the anomalous elasticity of bcc Fe–Cr and of the stiffening on spinodal decomposition. (a) Magnetic frustration: isolated Cr is antiparallel to Fe; Cr–Cr neighbours are frustrated and the Cr moments collapse (Quantum ESPRESSO SQS: $\langle m_{\mathrm{Cr}}\rangle = -0.58, -0.21, -0.03\,\mu_{\mathrm{B}}$ at 20, 35, 50% Cr), giving $\Delta G_{\mathrm{demix}} < 0$ and the spinodal window (QE squares, PWSCF–PAW model line). (b) Fermi surface of non-magnetic Cr (QE, $k_z = 0$): the electron jack at Γ nests onto the hole octahedron at H with $\mathbf{Q}_{\mathrm{SDW}}$; adding Fe (rigid band, $N(E_{\mathrm{F}})$ of Ref. [36]) resizes the two sheets in opposite senses and destroys the nesting (Fe makes the SDW of Cr commensurate at about 2.3 at.% and lowers T_{N} by 15–20 K per at.% [37]). (c) Fe-rich side: electronic topological transition of the majority-spin Fermi surface at 5–10% Cr [7,38] (schematic), and the Fermi-level DOS of the SQS. (d) The SDW is accompanied by a charge-density wave and a strain wave at $2\mathbf{Q}$ [37] (schematic); its lattice imprint is the softening of the longitudinal phonons towards $\mathbf{Q}$, seen in DFPT (non-magnetic Cr, $k = 24^3$) and in neutron scattering at 300 K [39]. (e) Decomposition places the Fe-rich α ($x = 0.12$) and Cr-rich α' ($x = 0.74$) phases of MD Fe–36Cr on stiffer parts of a convex $f_{\mathrm{SAW}}(x)$: the lever rule gives +4.0% with the MD (classical) curve, which accounts for most of the +5.1% obtained for the fully decomposed MD structure, and +7.7% with the EMTO-CPA (electronic) curve whose random-alloy minimum lies near Fe–36Cr. (f) Schematic of the TGS measurement over the MD microstructures (random and decomposed, real (001) planes). (g) (P)SAW frequencies of 10^4 grains at $\lambda = 6.4\,\mu$m: Fe–36Cr from MD (random, decomposed) and Fe–35Cr from QE; the QE tensor puts 11.7% of grains above the 609 MHz band edge.

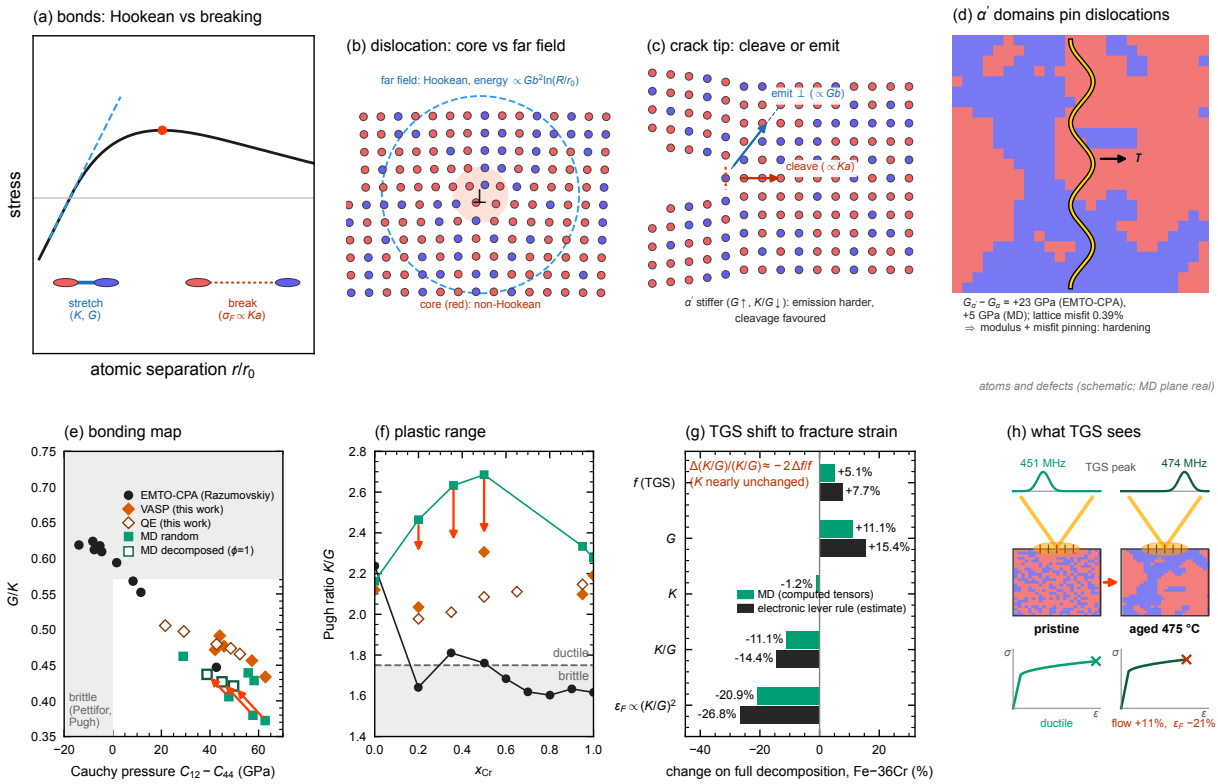


Figure S9: From elastic stiffening to embrittlement, after Pugh [4]. Top row, atoms and defects (schematic; the MD plane in (d) is the fully decomposed Fe–36Cr structure). (a) A bond stretches in the Hookean regime (slope $\propto K$, and G in shear) and breaks at the non-Hookean peak of the stress–separation curve, so fracture strength scales with Ka. (b) An edge dislocation in a random Fe–Cr lattice: its energy, $\propto Gb^2 \ln(R/r_0)$, is stored mostly in the Hookean far field, so the resistance to slip scales with Gb. (c) At a crack tip the alloy either cleaves the tip bond ($\propto Ka$) or emits a dislocation ($\propto Gb$); a stiffer Cr-rich α' makes emission harder and favours cleavage. (d) A dislocation bowing between α' domains: the shear-modulus misfit between α ($x = 0.12$) and α' ($x = 0.74$) is +23 GPa with the EMTO-CPA (electronic) moduli but only about +5 GPa with MD, and the lattice misfit is 0.39%, so modulus pinning is mainly an electronic-structure effect. Bottom row, data. (e) G/K against the Cauchy pressure $C_{12} - C_{44}$ [40] for all tensors; arrows: MD random → fully decomposed at 20, 36 and 50% Cr. (f) Pugh ratio K/G versus composition, with the empirical ductile/brittle boundary 1.75. (g) Fe–36Cr on full decomposition (MD): G rises by 11.1% while K changes by −1.2%, so K/G falls by 11.1%, i.e. $\Delta(K/G)/(K/G) \approx -2\,\Delta f/f$ with $\Delta f/f = +5.1\%$; with Pugh's $\varepsilon_F \propto (K/G)^2$ the fracture strain would fall by about 21% (27% for the electronic lever-rule estimate). (h) What TGS sees: pristine and aged Fe–36Cr (real MD planes) under the TGS spot. The measured peak moves from 451 to 474 MHz (MD), and the same 11% rise of G at nearly constant K raises the flow stress ($\propto Gb$) and, through $\varepsilon_F \propto (K/G)^2$, shortens the plastic range by about 21% (schematic stress–strain curves). The chain is: decomposition → Cr-rich α' domains → G +11%, K −1%, K/G −11% → slip harder, cleavage unchanged, and α' pins dislocations (electronic modulus misfit +23 GPa) → smaller plastic range, i.e. embrittlement; TGS reads the same G, with $\Delta(K/G)/(K/G) \approx -2\,\Delta f/f$. Pugh's relations are empirical correlations for pure metals; the numbers indicate direction and scale, not a prediction for CASS.

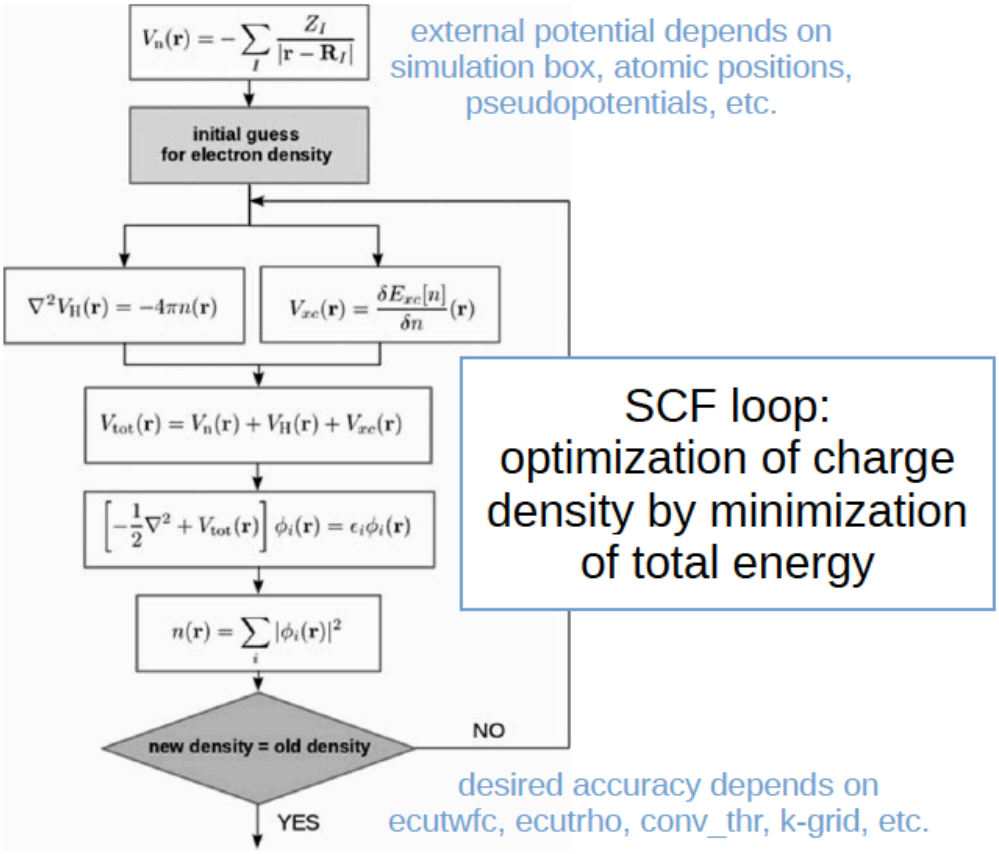


Figure S10: Self-consistent field (SCF) loop schematic illustrating self-consistent electronic-structure workflow [18].

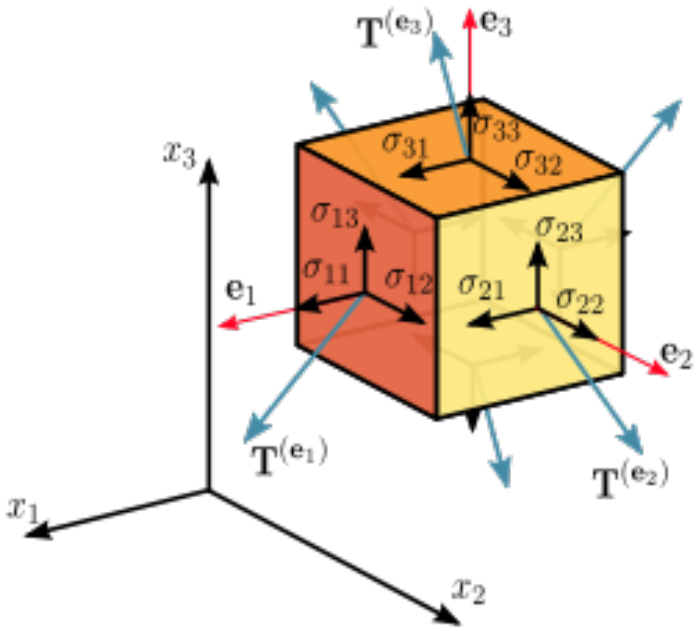
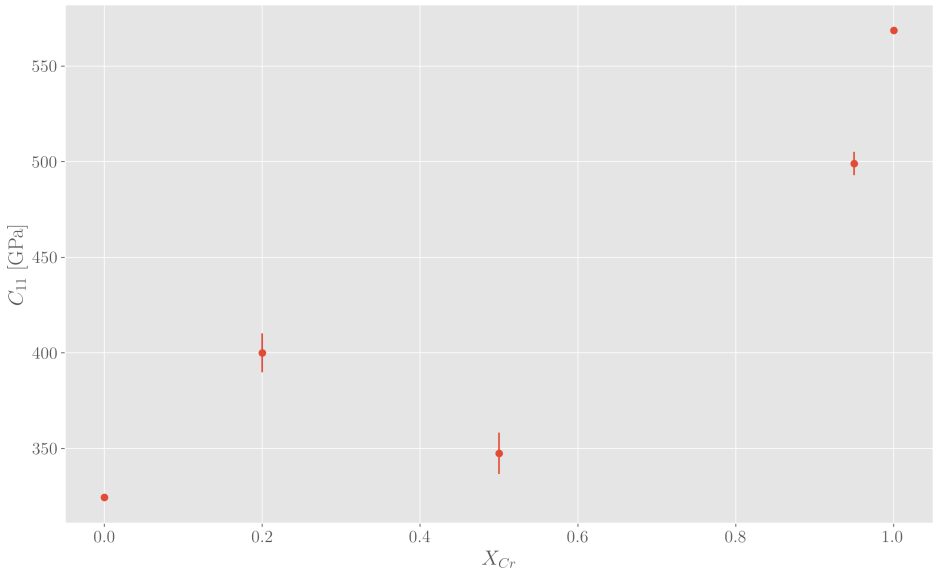
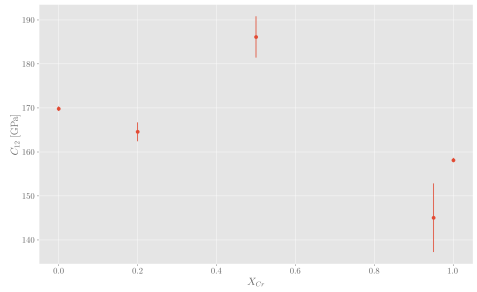


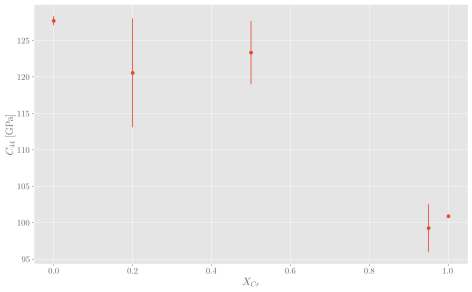
Figure S11: Stress tensor visualization [3].



(a) Normal constant, C_{11}, illustrating the longitudinal stiffness of bcc binary FeCr alloy as a function of chromium composition. This stiffness parameter is crucial for understanding the acoustic velocity and mechanical properties of the material.

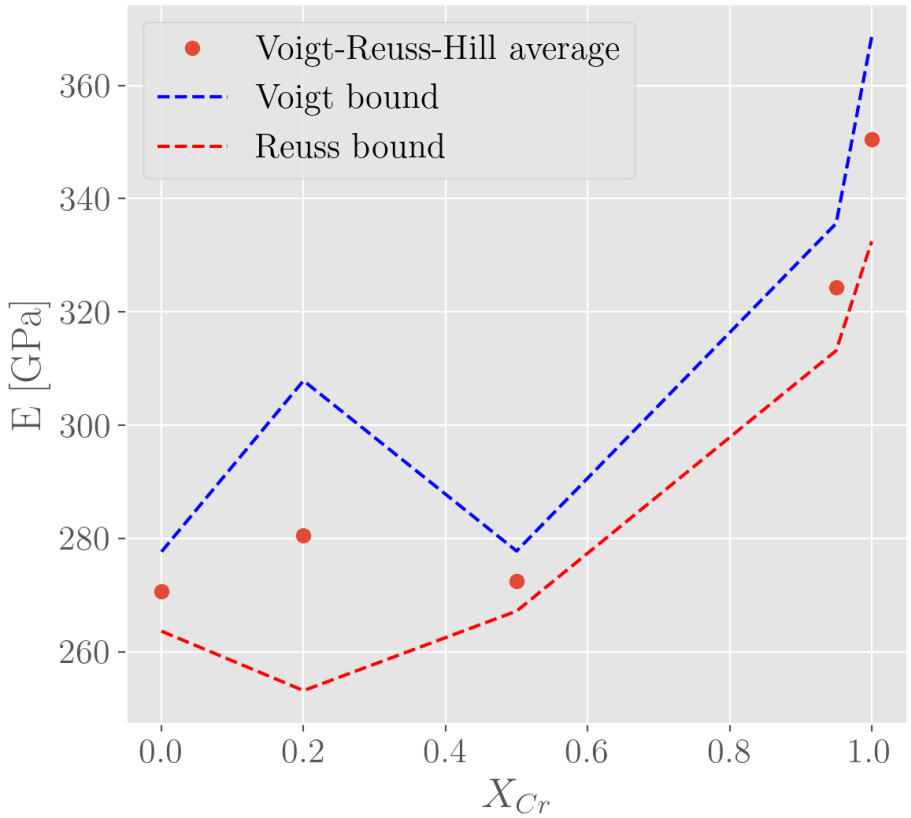


(b) The interplanar elastic constant C_{12}, showing the coupling between axial and lateral strains, indicative of the alloy's resistance to shear along crystallographic planes.

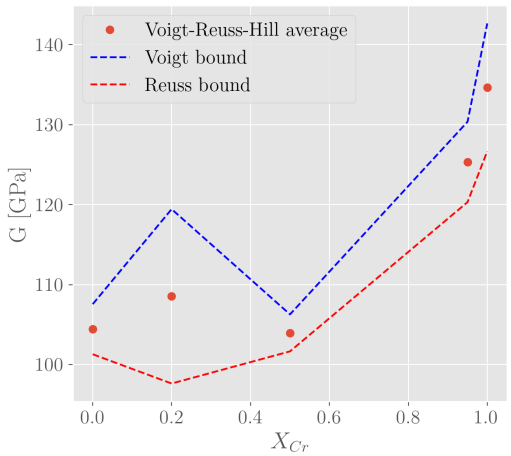


(c) Shear constant, C_{44}, representing shear stiffness against deformation without volume change, critical for assessing material's resistance to applied shear stress.

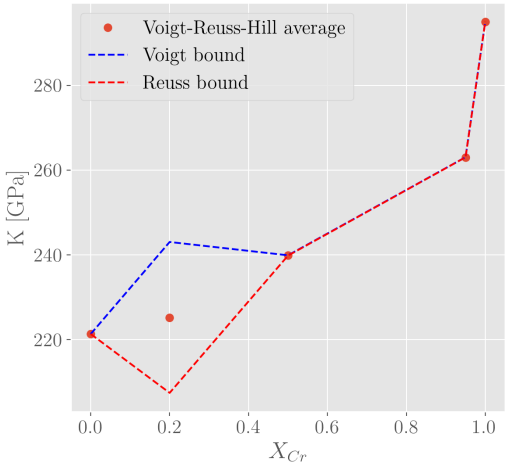
Figure S12: Elastic constants of homogeneous bcc FeCr from the present DFT calculations (40-atom SQS) as a function of Cr composition. (a) Shows the longitudinal stiffness parameter C_{11}, important for evaluating the material's response to compressive and tensile stresses. (b) Highlights C_{12}, which is essential for understanding the coupling between different types of strain. (c) Details C_{44}, crucial for evaluating the shear deformation behavior.



(a) Elastic (Young's) Modulus, E_{VRH}, indicating the material's stiffness and its ability to resist deformation under tensile stress.

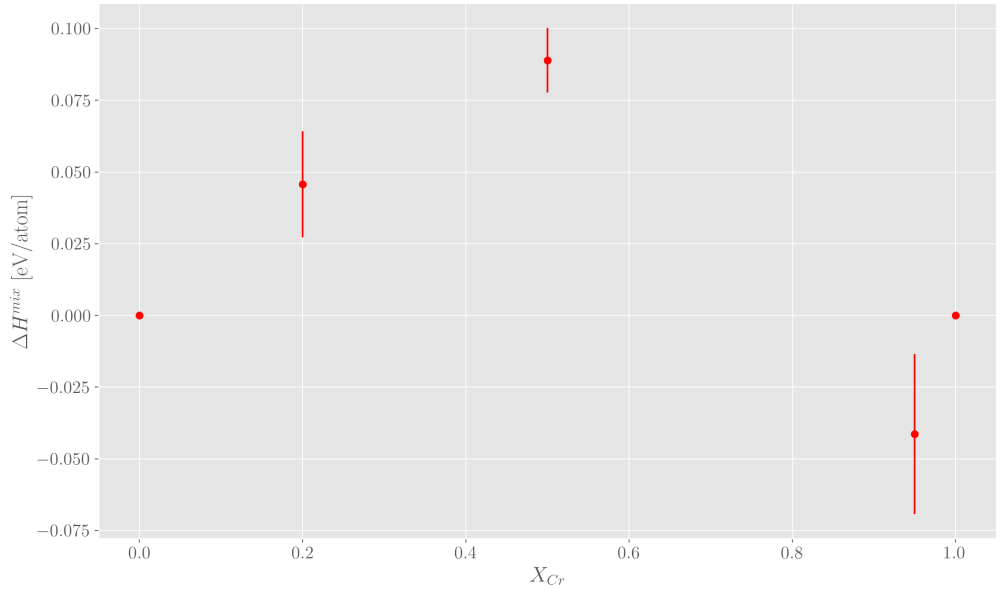


(b) Shear modulus, G_{VRH}, reflecting the material's ability to withstand shear deformation.

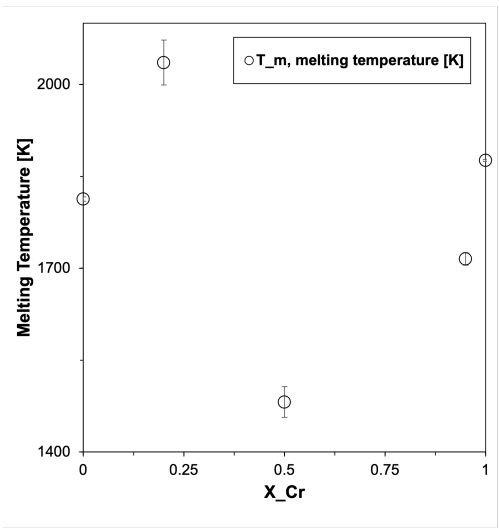


(c) Bulk modulus, K_{VRH}, which measures the material's resistance to uniform compression.

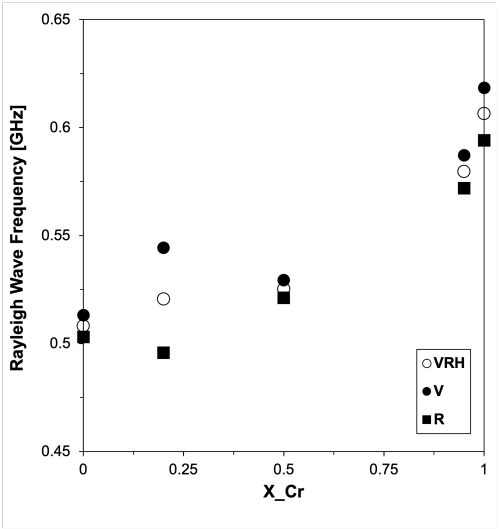
Figure S13: VRH elastic moduli of homogeneous bcc FeCr from the present DFT calculations as a function of Cr composition. (a) Elastic (Young's) Modulus demonstrates the material's overall stiffness. (b) Shear modulus is a measure of the alloy's rigidity against shear forces. (c) Bulk modulus assesses the material's compressibility under uniform pressure.



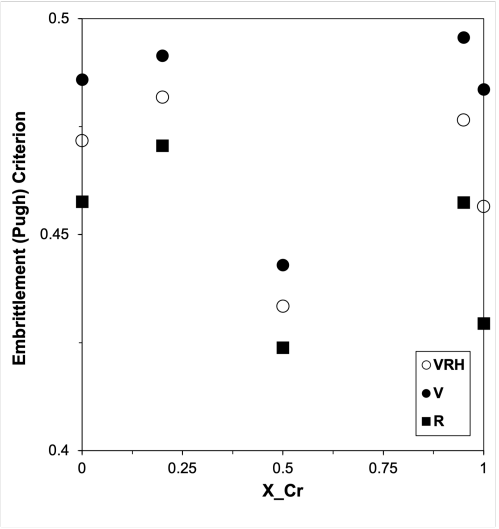
(a) Enthalpy of mixing, showing the thermodynamic favorability of phase separation in bcc δ-ferrite in cast austenitic stainless steels through spinodal decomposition.



(b) T_M, melting temperature, indicating how the thermal stability of the material changes with Cr composition.

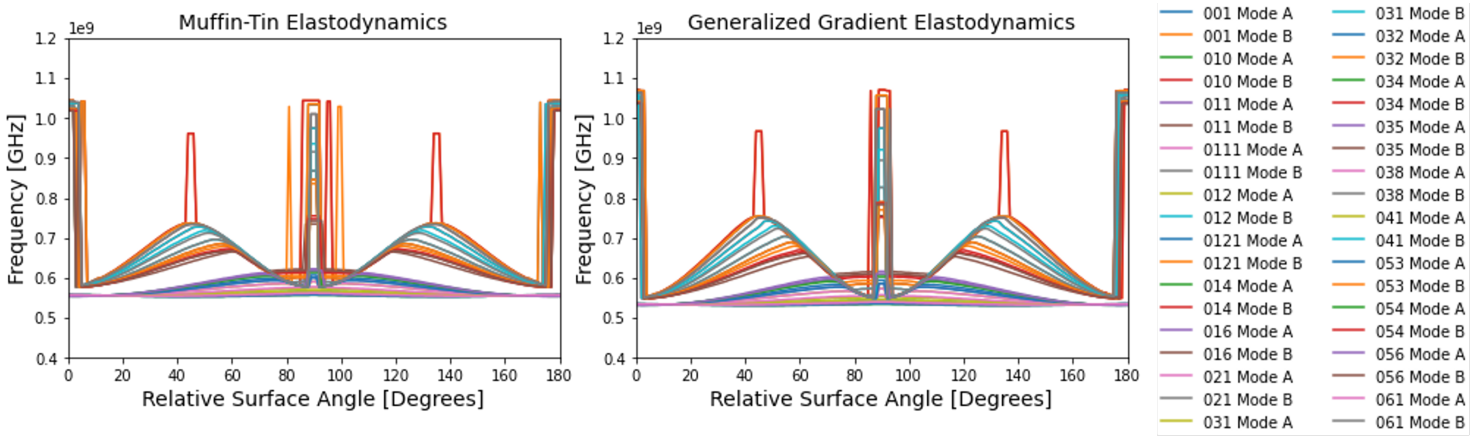


(c) Rayleigh wave frequencies, illustrating the impact of spinodal decomposition on the material's surface acoustic properties.

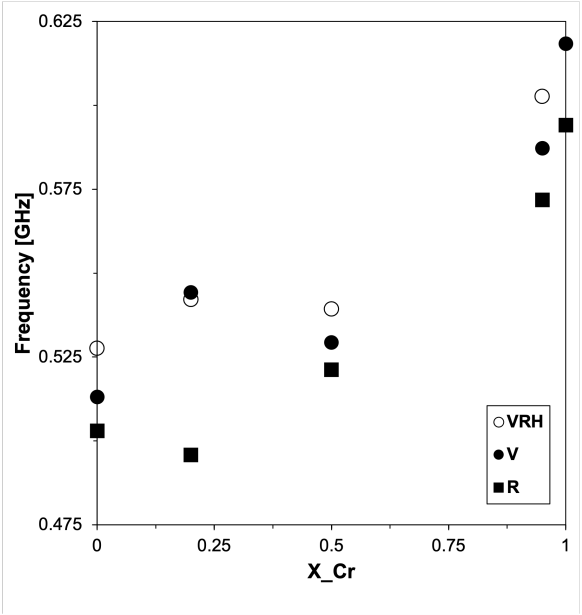


(d) Embrittlement (Pugh) Criterion, calculated to evaluate the embrittlement risk associated with varying Cr compositions.

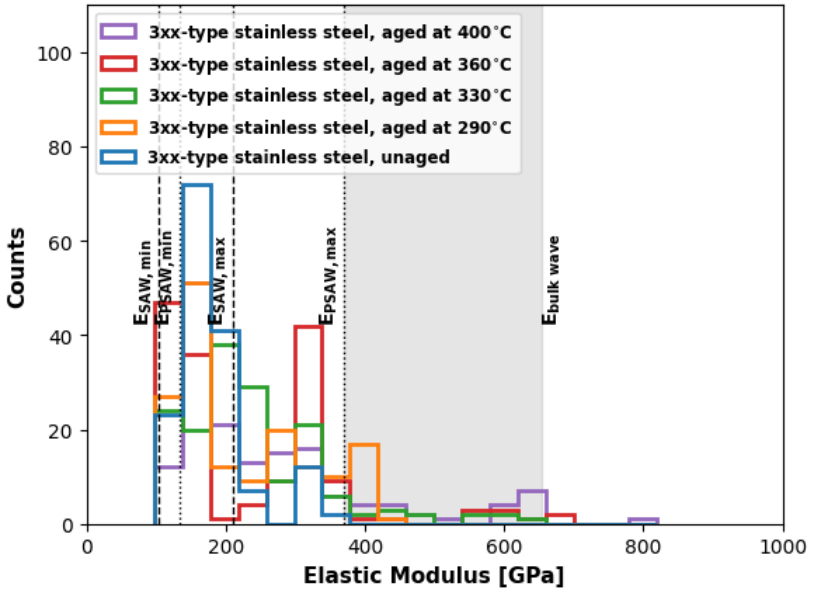
Figure S14: (a) The enthalpy of mixing demonstrates the thermodynamic drive towards phase separation indicative of spinodal decomposition, affecting the material's mechanical integrity. (b) Variations in the melting temperature as a function of Cr composition offer insights into the thermal stability and potential phase transformations within the alloy. (c) Changes in Rayleigh wave frequencies across compositions highlight alterations in surface acoustic properties, correlating with mechanical behavior. (d) The Embrittlement (Pugh) Criterion allows for assessing the risk of material embrittlement due to changes in composition, crucial for ensuring material reliability in critical applications.



(a) Predicted elastic frequencies showcasing the emergence of new modes within the elastodynamically-unallowed regions, signifying alterations in the material's mechanical properties due to spinodal decomposition.

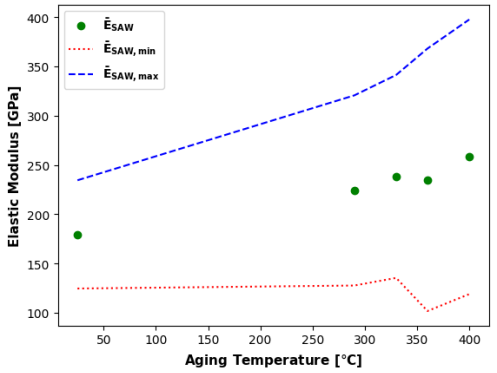


(b) Calculated acoustic frequencies across different Cr compositions, illustrating the material's acoustic response variability with composition. This graph provides a quantitative measure of the material's mechanical behavior and its dependence on the microstructural characteristics induced by varying chromium content.

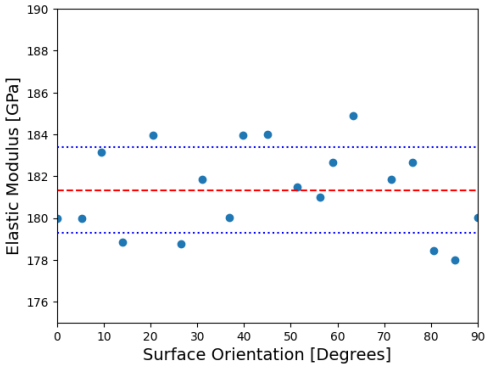


(c) Measured elastic moduli distribution from experimental data, highlighting the range and variability of material properties within stainless steel samples subjected to different thermal aging processes. The distribution underscores the significant impact of microstructural changes, such as spinodal decomposition, on the material's overall stiffness and mechanical integrity.

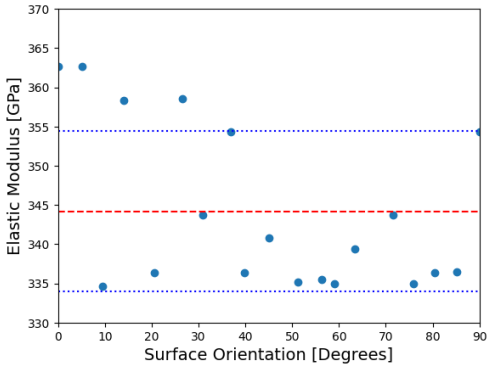
Figure S15: (a) Quantum elastodynamic frequencies obtained from density functional theory calculations reveal the complex interplay between microstructural evolution and mechanical properties in stainless steels undergoing spinodal decomposition. These findings are corroborated by (b) calculated acoustic frequencies that vary with Cr composition, providing a link between composition, microstructure, and acoustic properties. (c) The distribution of measured elastic moduli further validates the theoretical predictions, illustrating the broad impact of aging and compositional changes on the material's mechanical performance.



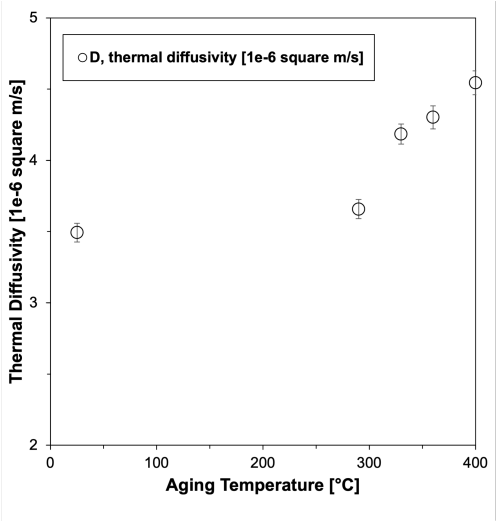
(a) Experimental elastic moduli of aged steels, displaying how aging temperature influences the material's stiffness. This graph illustrates the correlation between thermal treatment and the increase in elastic moduli, indicative of the material's response to microstructural changes over time.



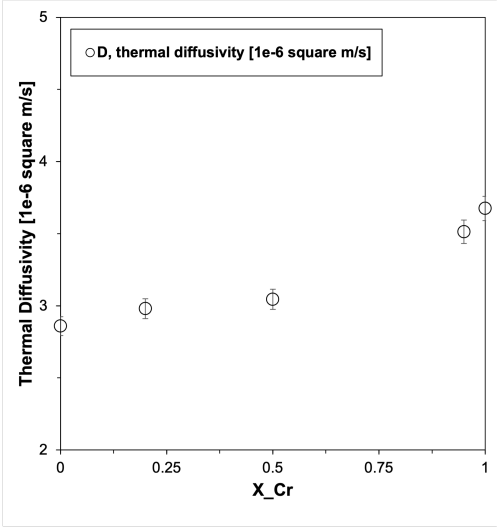
(b) Theoretical elastic moduli of pristine steels, offering a baseline comparison for understanding how spinodal decomposition affects material properties before and after aging. This comparison elucidates the stiffening effect of microstructural evolution on the alloy's mechanical behavior.



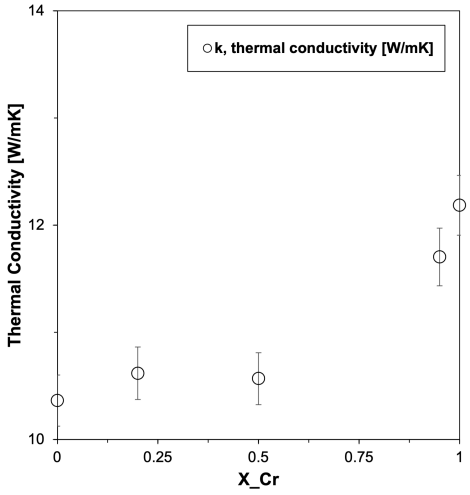
(c) Theoretical elastic moduli of spinodally decomposed steels, demonstrating the pronounced impact of microstructural evolution on the alloy's stiffness. This direct comparison to pristine conditions highlights the significant role of chromium composition and thermal aging in material performance.



(d) $D^{exp}_{thermal}$, experimental thermal diffusivity as measured in aged stainless steel, illustrating the thermal transport characteristics and their variation with temperature. This metric provides insights into the thermal stability and heat dissipation capabilities of the material post-aging.



(e) $D^{theoretical}_{thermal}$, theoretical thermal diffusivity indicating the expected thermal behavior based on composition and microstructural considerations. This predictive analysis aids in understanding the thermal management potential of the alloy across a range of compositions.



(f) $\kappa^{theoretical}_{thermal}$, thermal conductivity, showcasing the theoretical prediction of how efficiently the material conducts heat. This property is crucial for applications where thermal management is key, and it is influenced by the alloy's microstructure and composition.

Figure S16: Comprehensive analysis combining experimental and theoretical insights into the thermal and mechanical properties of stainless steels subjected to thermal aging. (a) Experimental elastic moduli trends with aging temperature highlight the stiffening effect of microstructural evolution. (b) and (c) Theoretical predictions for pristine and spinodally decomposed steels provide a comparative framework for assessing the impact of spinodal decomposition on material properties. (d) Experimental thermal diffusivity measurements, alongside (e) theoretical predictions for thermal diffusivity and (f) thermal conductivity, offer a holistic view of the material's thermal and mechanical behavior post-aging, emphasizing the interconnectedness of thermal properties, mechanical performance, and microstructural characteristics.